

Nifty 100 Financial Intelligence Platform

Institutional Fundamental Analyst Guide

1. Introduction & Architecture

Welcome to the Nifty 100 Financial Intelligence Platform Analyst Guide. This document serves as the core operational handbook for financial analysts consuming the API endpoints, inspecting the Streamlit dashboard, or downloading auto-compiled report books.

The platform operates on a **7-Layer Decoupled Architecture** designed to separate concerns, enforce strict database validations, and run performant financial computations. Data flows sequentially from L1 (Raw Excel Ingestion) to L3 (SQLite Database), L4 (KPI Ratio Engine), L5 (KMeans and NLP Analytics), L6 (ReportLab PDF Generation), and finally L7 (Interface Layer).

2. Operational Quickstart

Analysts can execute tasks in the workspace using the following commands defined in the project Makefile:

make load - Ingest raw data sheets and run 16 DQ rules.

make ratios - Populates computed ratios in the database.

make clustering - Fit KMeans, compute Z-scores and Pearson matrices.

make report - Rebuild all 92 tearsheets and sector PDFs.

make test - Execute the pytest suite (60 test cases).

make api - Start the FastAPI REST server on port 8000.

make dashboard - Run Streamlit app on port 8501.

3. Ratio Engine Formulas & Specifications

The L4 analytics layer processes 50+ financial ratios. The core definitions are:

3.1 Profitability & Return Ratios

Return on Equity (ROE): $\text{Net Profit} / \text{Total Equity (Equity Share Capital + Reserves)}$. If equity is negative, ROE returns 0.0 to prevent misleading math.

Return on Capital Employed (ROCE): $\text{EBIT} / \text{Capital Employed}$. Capital Employed is defined as $\text{Equity} + \text{Reserves} + \text{Long-term Borrowings}$.

Operating Profit Margin (OPM): $\text{Operating Profit} / \text{Revenue (Sales)}$. Missing bank operating margins are handled gracefully.

3.2 Solvency & Leverage Ratios

Debt-to-Equity (D/E): $\text{Borrowings} / \text{Total Equity}$. Enforces sector exclusions (e.g. D/E for banks/NBFCs is displayed with caution flags).

Interest Coverage Ratio (ICR): $(\text{Operating Profit} + \text{Other Income}) / \text{Interest Expense}$. When Interest is zero, the engine substitutes a score of **999.0**, which displays as 'Debt Free' in report summaries.

3.3 Cash Flow KPI Metrics

CFO Quality Score: $\text{Average CFO} / \text{PAT over 5 years}$. Ratios > 1.0 indicate 'High Quality Earnings'. Ratios < 0.5 flag 'Accrual Risk' (cash conversion lag).

CapEx Intensity: $\text{CapEx} / \text{Sales}$. Scaled as 'Asset-Light' ($< 3\%$) or 'Capital-Intensive' ($> 8\%$).

FCF Conversion: $\text{Free Cash Flow (CFO + CFI)} / \text{EBITDA}$.

4. Unsupervised Clustering & Outliers

The L5 intelligence layer segments the constituent universe to uncover statistical groupings without lookahead bias.

4.1 KMeans Cluster Profiles (K=5)

Using 5 standardized metrics (ROE, D/E, Revenue CAGR 5yr, FCF CAGR 5yr, OPM), companies are grouped into:

- 1. High-Quality Growth Compounder:** Low debt, high ROE (>18%), and strong double-digit growth.
- 2. Sturdy Cash Cows:** Highly profitable, asset-light, stable cash generation, low growth.
- 3. Leveraged / High Debt:** Debt-loaded companies, typical of capital-intensive utilities or NBFCs.
- 4. Steady Performers:** Near-average ROE, reasonable growth, standard leverage.
- 5. Underperformers / Cyclical Lows:** Muted growth, low ROE, or loss-making operations.

4.2 Sector Outlier Detection (Z-Score)

Outliers are flagged when a constituent company's key metrics deviate from its sector median by more than 3.0 standard deviations ($|Z| > 3.0$). This isolates statistical exceptions (e.g., extremely high leverage or exceptional ROE) for follow-up fundamental audit.

5. REST API Usage

FastAPI handles the API microservices. Start it using 'make api' and access the swagger page at <http://localhost:8000/>.

Common routes include:

GET /api/v1/companies - List constituents.

GET /api/v1/companies/{ticker}/ratios - Fetch computed ratios history.

GET /api/v1/screener?preset=quality_compounder - Run preset screen filter.

GET /api/v1/reports/tearsheet/{ticker} - Download the tearsheet PDF.